

LAB REPORT

Multi Node Clock Synchronization

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Course: Distributed Embedded Systems

Wien, 2026-06-09

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1 Introduction

This project was developed as part of the **Distributed Embedded Systems (DRS)** class.

1.1 Project Context and Objectives

The primary goal of this project is to build a Distributed Real-Time System (DRS) application capable of establishing a highly precise global time base across a dynamic cluster of hardware nodes (such as Raspberry Pis). The nodes communicate over mixed networks, including both Ethernet and Wi-Fi, utilizing UDP sockets in C/C++.

The overarching objective is to achieve a synchronization precision of $< 100 \mu\text{s}$ between the nodes. Crucially, the synchronization protocol is designed to run entirely in the user-space of a standard Linux OS without the need for custom kernel drivers.

1.2 Core Principles

The system architecture is guided by several core engineering requirements:

- **KISS Principle (Keep It Simple, Stupid):** The system requires zero manual configuration regarding network addresses or node roles. When a new node is plugged in, it integrates seamlessly.
- **Autonomous Discovery:** Nodes discover each other dynamically without a centralized registry.
- **Resilience:** The protocol dynamically filters latency jitter (especially asymmetrical jitter introduced by Wi-Fi) and is resilient to crash failures, ensuring there is no single point of failure if a leader node crashes.
- **Verifiability:** The claimed synchronization precision must be objectively measurable and falsifiable using external verification tools to prove physical simultaneity, rather than relying solely on internal software timestamps.

1.3 Authorship and Report Structure

It is important to note that both the codebase for this project and the generation of this report were heavily assisted by Artificial Intelligence (AI). The underlying system architecture and engineering requirements, however, were carefully thought out and designed by the architecture team.

The remainder of this report is structured to walk through these architectural decisions in detail. The subsequent chapters outline the complete system architecture—ranging from the hardware environment and network topology to the specific wire protocol and synchronization mathematics. Finally, the report concludes with an overview of the system’s telemetry data formatting and the development workflow used to build and verify the cluster.

2 CLAUDE.md

This file provides guidance to Claude Code (claude.ai/code) when working with code in this repository.

2.1 Project Overview

DRS (Distributed Real-time Synchronization) is a high-precision distributed clock synchronization system for a cluster of Raspberry Pi 4B nodes. The goal is to keep all nodes' virtual clocks aligned within $< 100 \mu\text{s}$, measured physically via GPIO pulses observed on a logic analyzer.

This is a **pure user-space C/C++ implementation** targeting **PREEMPT_RT Linux** on Raspberry Pi 4B (Cortex-A72). No kernel modules. No NTP/PTP/GPS. No external time authority. The cluster is self-synchronizing and internal-only.

The full architecture specification is in [drs_architecture_reviewed_fixed_v_21.md](#). The telemetry packet format is in [telemetry_format.md](#).

2.2 Architecture

2.2.1 Core Concept

Each node runs the same binary. One node is elected **leader** (lowest NodeID wins via modified Bully algorithm). Followers exchange timestamps with the leader using a PTP-style 4-timestamp exchange (T1–T4) over UDP multicast on a single-hop Gigabit Ethernet LAN.

Synchronization adjusts a **virtual clock layer only** — the host OS clock (CLOCK_REALTIME) is never touched. All timestamps use CLOCK_MONOTONIC_RAW.

2.2.2 Virtual Clock

$$T_{\text{global}} = (T_{\text{local_raw}} \times \text{Rate}) + \text{Offset} - \text{LatencyCorrection_ns}$$

- Rate is Q32.32 fixed-point (no floating-point in hot path)
- Offset is phase correction in nanoseconds
- LatencyCorrection_ns is measured once per calibration phase

2.2.3 Node State Machine

GROUND → CALIBRATION → LISTEN → CANDIDATE → LEADER or FOLLOWER

Fallback: HOLDOVER (freerun for up to 10 s when leader is lost), then re-election.

2.2.4 Synchronization Loop

- Runs exclusively on **CPU Core 3** (isolated via `isolcpus=3 nohz_full=3 rcu_nocbs=3`)
- `SCHED_FIFO` priority 85
- 50 ms exchange period, 100 ms leader heartbeat, 3-heartbeat timeout
- Uses `clock_nanosleep()` for yielding (prevents softirq starvation/SSH lockout)
- `mlockall(MCL_CURRENT | MCL_FUTURE)` — no page faults in hot path

2.2.5 Control Algorithm

Dual-loop PI controller:

$$\text{Rate_new} = \text{Rate_old} + K_p \cdot \theta + K_i \cdot \int \theta dt$$

Defaults: $K_p=0.05$, $K_i=0.005$. Max slew ± 1000 ppm. Integrator clamped to ± 1000 ppm.

Step mode (hard phase correction) requires 3 consecutive samples confirming $|\theta| > 1$ ms.

2.2.6 Min-Delay Filter

RTT window size = 10. Only samples within $\text{min_RTT} + 10 \mu\text{s}$ are accepted to reject scheduler spikes and interrupt bursts.

2.3 Wire Protocol

- Fixed 64-byte UDP packets, Big Endian
- Magic: 0x44525354 (“DRST”), Version 1
- Message types: `ANNOUNCE` (0x01), `SYNC_REQ` (0x02), `SYNC_RESP` (0x03)
- CRC32 with IEEE 802.3 polynomial, covers full 64 bytes including zero-padded reserved fields
- All fields serialized explicitly with `htons()/htonl()/explicit 64-bit conversion` — **no raw struct casting**
- Timestamps captured via `S0_TIMESTAMPING` (hardware NIC preferred, kernel software fallback)

2.3.1 Network

- Multicast group: 239.192.88.100, port 47200
- `IP_MULTICAST_LOOP` = 0 (suppress sender-local loopback only)
- Node IPs: 10.0.0.XY where X=team ID, Y=node ID

2.4 GPIO

- **GPIO 18**: 10 ms HIGH pulse once per global second (rising edge = second boundary). Jitter < 20 μs .
- **GPIO 23**: Sync health indicator. LOW = stable ($|\text{offset}| < 100 \mu\text{s}$ for 10 continuous seconds). HIGH = any other state.

2.5 Hard Constraints (Never Violate)

- **Never** call `clock_settime()`, `adjtimex()`, or step `CLOCK_REALTIME`
 - **Never** use floating-point in the synchronization hot path
 - **Never** allocate heap memory in the hot path
 - **Never** do filesystem/console I/O in the sync loop
 - **Never** use TCP for synchronization transport
 - **Never** allow WLAN interrupts or Ethernet IRQs to execute on Core 3
 - **Never** use mutex blocking in the sync loop — use atomic acquire/release for Rate/Offset updates
-

2.6 Deployment

The binary deploys as a systemd service (`/usr/local/bin/drs_sync`) with `CPUAffinity=3`, `CPUSchedulingPolicy=fifo`, `CPUSchedulingPriority=85`, `LimitMEMLOCK=infinity`.

Disable before running: `systemd-timesyncd`, `chronyd`, `ntpd`, `ptp4l`, `phc2sys`.

CPU governor must be set to performance.

2.7 Think Before Coding

Don't assume. Don't hide confusion. Surface tradeoffs.

Before implementing:

- State your assumptions explicitly. If uncertain, ask.
 - If multiple interpretations exist, present them - don't pick silently.
 - If a simpler approach exists, say so. Push back when warranted.
 - If something is unclear, stop. Name what's confusing. Ask.
-

2.8 Simplicity First

Minimum code that solves the problem. Nothing speculative.

- No features beyond what was asked.
- No abstractions for single-use code.
- No “flexibility” or “configurability” that wasn't requested.
- No error handling for impossible scenarios.
- If you write 200 lines and it could be 50, rewrite it.

Ask yourself: “Would a senior engineer say this is overcomplicated?” If yes, simplify.

2.9 Surgical Changes

Touch only what you must. Clean up only your own mess.

When editing existing code:

- Don't "improve" adjacent code, comments, or formatting.
- Don't refactor things that aren't broken.
- Match existing style, even if you'd do it differently.
- If you notice unrelated dead code, mention it - don't delete it.

When your changes create orphans:

- Remove imports/variables/functions that YOUR changes made unused.
- Don't remove pre-existing dead code unless asked.

The test: Every changed line should trace directly to the user's request.

2.10 Goal-Driven Execution

Define success criteria. Loop until verified.

Transform tasks into verifiable goals:

- "Add validation" → "Write tests for invalid inputs, then make them pass"
- "Fix the bug" → "Write a test that reproduces it, then make it pass"
- "Refactor X" → "Ensure tests pass before and after"

For multi-step tasks, state a brief plan:

1. [Step] → verify: [check]
2. [Step] → verify: [check]
3. [Step] → verify: [check]

Strong success criteria let you loop independently. Weak criteria ("make it work") require constant clarification.

2.11 Git Style

Commit proactively after each plan phase is complete, tested, and confirmed working — do not wait to be asked. Follow the git style: one file (or tightly related file group) per commit. Never bundle unrelated files into a single commit. Keep commit messages short and focused — one change, one message. Do not create long commit message chains.

Do not touch already-committed code unless the task requires it. No reformatting, no comment tweaks, no whitespace cleanup as a side effect. Before committing, review the full diff and remove any unintended changes.

Never commit “current state” snapshots or plan markdown files. These exist as working files only and must not enter git history. Before every commit, review what is staged and exclude any planning or status documents.

Autonomous commits (made during implementation without being asked): Claude is the sole author — do not add a co-author trailer.

User-requested commits (user explicitly asks to commit): Add Claude as co-author:

Co-Authored-By: Claude Sonnet 4.6 <noreply@anthropic.com>

Master Architecture Document

DRS High-Precision Distributed Synchronization Cluster

Consolidated Version 2.1 (Reviewed & Clarified)

May 14, 2026

1. Project Definition

1.1 Mission Statement

The DRS project establishes a deterministic, high-precision distributed global time base across a dynamic cluster of Raspberry Pi nodes operating exclusively in Linux user space.

The system shall maintain a physical synchronization delta of:

$\Delta t_{\text{physical}} \leq 100 \mu\text{s}$

as measured externally using GPIO-generated synchronization pulses observed on a logic analyzer or oscilloscope.

The architecture prioritizes:

- deterministic behavior
- operational simplicity
- fail-safe recovery
- real-time scheduling integrity
- minimal protocol complexity
- bounded timing behavior

1.2 Core Design Philosophy

The synchronization subsystem SHALL follow the KISS principle:

Keep It Simple, Stupid (KISS)

The protocol intentionally avoids:

- distributed consensus frameworks
- dynamic spanning trees
- kernel modifications
- external time authorities
- heavyweight serialization
- multi-hop routing
- dynamic routing protocols
- cryptographic processing in the synchronization hot path

The synchronization cluster is designed exclusively for:

- single-hop Layer-2 Gigabit Ethernet
- trusted laboratory environments
- fail-stop node behavior
- deterministic LAN conditions

1.3 Revised Core Constraints

- Wired Gigabit Ethernet SHALL be the exclusive synchronization transport.
- WLAN MAY be used exclusively for out-of-band management and provisioning.
- CPU Core 3 SHALL be reserved exclusively for synchronization operations.
- WLAN interrupts SHALL NEVER execute on Core 3.
- Ethernet IRQs SHALL NEVER execute on Core 3.
- The Linux host system clock SHALL NEVER be modified.
- All synchronization adjustments SHALL occur exclusively inside the virtual clock layer.

Explicitly prohibited:

- `clock_settime()`
- `adjtimex()`
- `CLOCK_REALTIME` stepping
- kernel PLL interaction

1.4 Synchronization Paradigm

The system implements:

Internal Monotonic Synchronization

A single elected leader defines the cluster reference timeline.

Follower nodes estimate:

- phase offset
- frequency drift
- path delay

relative to the elected leader.

The synchronization domain is entirely internal.

The cluster SHALL NOT synchronize to:

- UTC
- NTP
- GPS
- PTP grandmasters
- wall-clock time

1.5 Virtual Clock Model

The virtual clock mapping SHALL be defined as:

$$T_{\text{global}} = (T_{\text{local_raw}} \times \text{Rate}) + \text{Offset} - \text{LatencyCorrection_ns}$$

Where:

Parameter	Description
$T_{\text{local_raw}}$	CLOCK_MONOTONIC_RAW timestamp
Rate	Fixed-point frequency scaling multiplier
Offset	Phase correction in nanoseconds
LatencyCorrection_ns	Static calibration compensation

Rate Representation

Rate SHALL be represented as signed 64-bit Q32.32 fixed-point.

Definitions:

Value	Meaning
0x00000001_00000000	1.0x nominal clock rate
+1000 ppm	Maximum positive slew
-1000 ppm	Maximum negative slew

Floating-point arithmetic SHALL NOT be used inside the synchronization hot path.

1.6 Time Base Definition

The synchronization engine SHALL operate exclusively on:

CLOCK_MONOTONIC_RAW

All protocol timestamps SHALL represent:

Absolute nanoseconds since local kernel boot.

The cluster timeline SHALL be defined relative to the elected leader's monotonic boot-time origin.

The system SHALL NOT use:

- Unix Epoch
- wall-clock time
- CLOCK_REALTIME

Optional wall-clock export layers MAY exist externally but remain outside synchronization scope.

2. Hardware & Operating Environment

2.1 Supported Hardware

Target platform:

- Raspberry Pi 4B
- Quad-core Cortex-A72
- Gigabit Ethernet
- PREEMPT_RT Linux kernel

Recommended:

- passive cooling
- fixed thermal conditions
- wired Ethernet only

2.2 GPIO Hardware Interface

GPIO 18 — Synchronization Pulse

GPIO 18 SHALL generate:

- a 10 ms HIGH pulse
- once per global second

The rising edge SHALL define the global second boundary.

The pulse SHALL remain active in all operational states except:

GROUND

Timing Constraints

| Property | Requirement | |---| | Max pulse edge jitter | <20 μ s | | Pulse source | Virtual global clock only | | Pulse generation thread | RT synchronization thread |

GPIO 23 — Synchronization Health Indicator

GPIO 23 SHALL indicate synchronization health.

| State | Meaning | |---| | LOW | Stable synchronized operation | | HIGH | Startup, degraded sync, recovery, calibration, or fault |

GPIO 23 SHALL transition LOW only after:

|Offset| < 100 μ s

for a continuous:

10-second interval

2.3 External Verification

Recommended instrumentation:

- multi-channel digital oscilloscope
- logic analyzer

Minimum sample rate:

≥ 1 MHz

Measurement criteria:

- physical delta between GPIO 18 rising edges

2.4 Kernel & OS Hardening

Required Kernel Parameters

```
isolcpus=3  
nohz_full=3  
rcu_nocbs=3
```

Scheduling Requirements

Synchronization thread SHALL:

- execute exclusively on Core 3
- use SCHED_FIFO
- use priority 85

CPU Frequency Requirements

CPU governor SHALL be:

performance

Turbo and dynamic voltage scaling SHOULD be disabled.

Memory Requirements

All synchronization process memory SHALL be locked using:

```
mlockall(MCL_CURRENT | MCL_FUTURE)
```

No page faults SHALL occur in the synchronization hot path.

Time Service Isolation

The following services SHALL be disabled:

- systemd-timesyncd
- chronyd
- ntpd
- ptp4l
- phc2sys

3. Network Architecture

3.1 Network Topology

Topology assumptions:

- single-hop Ethernet
- switched Gigabit LAN
- no routing
- no NAT
- no Wi-Fi bridges
- standard MTU 1500
- no jumbo frames

3.2 IP Addressing

Static schema:

10.0.0.XY

| Field | Meaning | |---| | X | Team ID (1-8) | | Y | Node ID (1-3) |

Example:

Team 4 Node 2 → 10.0.0.42

3.3 Discovery Mechanism

Discovery SHALL use UDP multicast.

| Parameter | Value | |---| | Group | 239.192.88.100 | | Port | 47200 | | IP_MULTICAST_LOOP | 0 |

IP_MULTICAST_LOOP = 0 SHALL disable sender-local multicast loopback only.

3.4 Security Assumptions

The synchronization protocol assumes:

- trusted LAN operation
- fail-stop node behavior
- bounded congestion
- non-malicious traffic

The protocol provides NO:

- authentication
- encryption
- replay protection
- Byzantine fault tolerance
- confidentiality guarantees

Deployment outside trusted laboratory environments is NOT supported.

4. DRS Wire Protocol

4.1 Packet Format

All packets SHALL:

- use fixed-size binary encoding
- use network byte order (Big Endian)
- avoid dynamic allocation in the hot path
- avoid text serialization

Raw struct casting is prohibited.

All fields SHALL be serialized explicitly using:

- `htons()`
- `htonl()`
- explicit 64-bit endian conversion

4.2 Packet Size

All packets SHALL be exactly:

64 bytes

4.3 Packet Layout

Field	Type	Size	Description
Magic	uint32	4	Constant: 0x44525354 ("DRST")
Version	uint8	1	Protocol version
MsgType	uint8	1	Message type
Flags	uint8	1	Synchronization state flags
Reserved	uint8	1	Reserved
Seq	uint16	2	Rolling sequence number
NodeID	uint32	4	Unique node ID
ElectionTerm	uint32	4	Leader election epoch
T1	uint64	8	Follower transmit timestamp
T2	uint64	8	Leader receive timestamp
T3	uint64	8	Leader transmit timestamp
T4	uint64	8	Follower receive timestamp
CRC32	uint32	4	Packet integrity
Padding	uint8[12]	12	Reserved
Total	—	64	Total packet size

4.4 Message Types

Value	Meaning
0x01	ANNOUNCE
0x02	SYNC_REQ
0x03	SYNC_RESP

4.5 Flags Field

Bit	Meaning
0	LEADER
1	HOLDOVER
2	CALIBRATED
3	FAULT
4-7	Reserved

Reserved bits SHALL be zero.

4.6 Timestamp Semantics

| Timestamp | Meaning | |---| | T1 | Sync request send time | | T2 | Kernel-captured receive timestamp
| | T3 | Kernel-captured transmit timestamp | | T4 | Sync response receive timestamp |

All timestamps SHALL use:

CLOCK_MONOTONIC_RAW nanoseconds

4.7 Timestamping Mechanism

Mandatory Linux API:

SO_TIMESTAMPING

Accepted modes:

1. hardware NIC timestamping
2. kernel software timestamping

The highest precision mode supported by the platform SHALL be selected automatically.

4.8 CRC32

CRC32 SHALL use:

- IEEE 802.3 polynomial 0x04C11DB7
- initial value 0xFFFFFFFF
- reflected input and output

Reserved padding bytes SHALL:

- be zero-filled
- be included in CRC validation

5. Synchronization Mathematics & Control

5.1 Offset Estimation

$$\theta = ((T2 - T1) + (T3 - T4)) / 2$$

5.2 Round Trip Delay

$$\delta = (T4 - T1) - (T3 - T2)$$

5.3 Min-Delay Filter

The synchronization engine SHALL maintain:

- rolling RTT window size = 10

Only samples within:

$$\text{minimum RTT} + 10 \mu\text{s}$$

SHALL be accepted.

Rejected samples include:

- scheduler spikes
- interrupt storms
- retransmission artifacts
- congestion bursts

5.4 Automated Calibration

Each node SHALL perform self-latency calibration before entering synchronization states.

Calibration SHALL:

- use SO_TIMESTAMPING loopback
- use minimum 50 samples
- use minimum-delay selection
- reject outliers >20 μs from minimum

Calibration output SHALL define:

LatencyCorrection_ns

Calibration SHALL re-run:

- during startup
- after leader election
- after HOLDOVER expiration
- after synchronization fault recovery

5.5 Dual-Loop PI Clock Discipline

Step Mode

A hard phase step SHALL require:

3 consecutive samples confirming $|\theta| > 1 \text{ ms}$

Only the virtual clock MAY step.

The host OS clock SHALL NEVER be modified.

Slew Mode

If:

$|\theta| \leq 1 \text{ ms}$

then gradual frequency correction SHALL occur through PI control.

5.6 PI Controller

$\text{Rate_new} = \text{Rate_old} + K_p \cdot \theta + K_i \cdot \int \theta dt$

Recommended defaults:

| Parameter | Value | |---| | Kp | 0.05 | | Ki | 0.005 | | Update Period | 50 ms |

Integrator SHALL be clamped to:

$\pm 1000 \text{ ppm equivalent}$

5.7 Slew Rate Limit

Maximum correction rate:

1000 ppm

Equivalent to:

50 μ s per 50 ms cycle

6. Leader Election & State Machine

6.1 Election Model

The system implements a modified Bully algorithm.

Lowest NodeID SHALL win leadership.

Election timeout SHALL be randomized within:

250–500 ms

to reduce election collisions.

6.2 Node States

| State | Description | |---| | GROUND | Startup stabilization | | CALIBRATION | Self-latency measurement | | LISTEN | Passive discovery | | CANDIDATE | Election in progress | | FOLLOWER | Synchronized to leader | | LEADER | Authoritative clock source | | HOLDOVER | Temporary freerun mode | |

6.3 GROUND State

After boot:

- filters SHALL be zeroed
- transmissions SHALL be suppressed
- multicast listening SHALL remain active

Duration:

2 seconds

6.4 Heartbeats

| Property | Value | |---| | Leader announce interval | 100 ms | | Follower timeout | 3 missed heartbeats | | Sync exchange period | 50 ms |

6.5 Immediate Demotion

If a lower-ID node appears:

- current leader SHALL demote immediately
- election SHALL restart deterministically

The lower-ID node SHALL assume leadership only after:

- successful calibration
- valid heartbeat transmission

6.6 HOLDOVER Mode

If leader communication is lost:

- node SHALL enter HOLDOVER
- last stable Rate SHALL be maintained
- Offset SHALL be frozen
- GPIO 23 SHALL transition HIGH
- drift adaptation SHALL stop

Maximum HOLDOVER duration:

10 seconds

After expiration:

- node SHALL re-enter election

7. Real-Time Scheduling

Synchronization thread SHALL:

- use SCHED_FIFO
- use priority 85
- remain pinned to Core 3

The synchronization loop MUST periodically yield using:

```
clock_nanosleep()
```

to prevent:

- softirq starvation
- scheduler deadlock
- SSH lockout

8. Failure Recovery

8.1 Retry Storm Protection

Failed nodes SHALL reduce multicast rate to:

5 Hz

after:

5 consecutive synchronization failures

8.2 Filter Reset Conditions

Synchronization filters SHALL reset on:

- reboot
- sequence discontinuity
- leader change
- timestamp overflow
- phase correction >5 ms

8.3 Sequence Wraparound

Sequence numbers SHALL use unsigned 16-bit rollover semantics.

A sequence transition:

65535 → 0

SHALL be treated as valid wraparound.

Backward jumps exceeding:

1024 sequence values

SHALL invalidate synchronization state.

9. Lock-Free Clock Access

The virtual clock SHALL support lock-free concurrent readers.

The synchronization thread SHALL be the exclusive writer.

Reader access SHALL:

- avoid mutex blocking
- avoid dynamic allocation
- guarantee monotonic reads

Offset and Rate updates SHALL use:

- atomic 64-bit operations
- acquire/release memory ordering

10. Functional Requirements

| ID | Requirement | |—| | F-1 | Autonomous multicast discovery | | F-2 | Virtual global monotonic clock
| | F-3 | Min-delay filtering | | F-4 | Deterministic bully election | | F-5 | Explicit fault recovery mapping
| | F-6 | Lock-free telemetry | | F-7 | Stable convergence detection | | F-8 | Immediate leader demotion |
| F-9 | Non-blocking writer precedence | | F-10 | Monotonic nanosecond representation | | F-11 | GPIO
validation pulse | | F-12 | Fixed performance governor | | F-13 | SO_TIMESTAMPING support | | F-14 |
Automatic filter reset | | F-15 | Fixed 64-byte packets | | F-16 | 100 ms synchronization timeout | | F-17
| Step vs slew discipline | | F-18 | Fail-silent GPIO indicator | | F-19 | Seamless late joiners | | F-20 |
Deterministic isolated startup | | F-21 | Static latency calibration | | F-22 | Retry-storm suppression | |
F-23 | IRQ affinity isolation | | F-24 | Explicit endian-safe serialization | | F-25 | Holdover freerun support
| | F-26 | CRC packet validation | | F-27 | Sequence wraparound handling | | F-28 | Integral windup
prevention | | F-29 | Step confirmation hysteresis | | F-30 | Convergence signaling | | F-31 | Automated
calibration | | F-32 | Calibration-on-election |

11. Non-Functional Requirements

| ID | Requirement | |—| | NF-1 | Physical pulse delta <100 μ s | | NF-2 | Pure user-space implementation | | NF-3 | Jitter resilience under mixed traffic | | NF-4 | Lock convergence within 10 s | | NF-5 | 3-heartbeat hysteresis | | NF-6 | Writer never blocked by readers | | NF-7 | SCHED_FIFO priority 85 | | NF-8 | mlockall memory locking | | NF-9 | No hot-path disk or console I/O | | NF-10 | Isolated Core 3 execution | | NF-11 | Max slew = 1000 ppm | | NF-12 | No dynamic heap allocation in hot path | | NF-13 | No packet fragmentation | | NF-14 | Deterministic packet parsing | | NF-15 | No mutex contention in sync loop | | NF-16 | Core 3 sanctity |

12. systemd Deployment

Example:

```
[Unit]
Description=DRS Synchronization Service
After=network-online.target
Wants=network-online.target

[Service]
Type=simple
ExecStart=/usr/local/bin/drs_sync
Restart=on-failure
RestartSec=5s
CPUAffinity=3
CPUSchedulingPolicy=fifo
CPUSchedulingPriority=85
LimitRTPRIO=95
LimitMEMLOCK=infinity
NoNewPrivileges=true

[Install]
WantedBy=multi-user.target
```

13. Performance Targets

Component	Budget	Actual
NIC timestamp jitter	<20 μ s	
Scheduler latency	<30 μ s	
Filter residual	<20 μ s	
GPIO generation	<20 μ s	
Total expected worst-case	<100 μ s	

14. Architectural Constraints

The following are explicitly prohibited:

- modifying kernel clocks
- custom kernel modules
- TCP transport
- Wi-Fi synchronization operation
- distributed consensus protocols
- mutex blocking in hot path
- filesystem I/O in synchronization loop
- floating-point exceptions in RT thread

15. Final System Goal

The DRS cluster SHALL behave as a single deterministic distributed timing domain where all participating nodes produce externally measurable synchronization pulses aligned within:

$<100\ \mu\text{s}$

under:

- PREEMPT_RT Linux
- standard user-space execution
- single-hop Ethernet
- dynamic node membership
- moderate background traffic conditions

3 DRS Telemetry Requirements

Version: 1.0

Date: 2026-06-01

Applies to: All cluster nodes running `drs_sync`

3.1 Overview

Every node emits telemetry records over UDP unicast to a designated receiver (e.g. a logging machine or visualizer). The telemetry stream is fire-and-forget — packet loss is acceptable. The sync hot path is never blocked by telemetry.

3.2 Transport

| Parameter | Value | |----|----| | Protocol | UDP unicast | | Destination port | **4242** | | Destination IP | Passed as the second CLI argument to `drs_sync` (e.g. `10.0.0.1`) | | Source | Any ephemeral port assigned by the OS | | Byte order | **Little-endian** (native RPi/x86 byte order — no byte-swapping) |

If no destination IP is provided, the binary defaults to `127.0.0.1` (loopback only).

3.3 Packet Format

Each packet is exactly **40 bytes**, binary, with no framing or length prefix.

| Offset | Size (B) | C type | Field | Description | |----|-----|-----|-----|-----| | 0 | 8 | int64 | timestamp_ns | CLOCK_MONOTONIC_RAW on the sending node, nanoseconds | | 8 | 4 | int32 | state | Node state enum value (see [State Values](#)) | | 12 | 8 | int64 | offset_ns | Measured clock offset from leader, nanoseconds. **0 for leader packets.** | | 20 | 8 | int64 | rtt_ns | Round-trip time of the sync exchange, nanoseconds. **0 for leader packets.** | | 28 | 8 | int64 | rate_q32 | Virtual clock rate as Q32.32 fixed-point (see [Rate Field](#)) | | 36 | 4 | uint32 | node_id | Sender's node ID (matches last octet of its IP: `10.0.0.<node_id>`) |

All fields are stored in native little-endian byte order. There is no magic number, version field, or CRC in the telemetry packet.

3.4 State Values

The state field carries the current node state machine value at the time the record is emitted.

Value	Name	Meaning
0	GROUND	Startup hold-off (~2 s), not yet participating
1	CALIBRATION	Measuring self-latency; one-shot at startup
2	LISTEN	Waiting to hear a leader before entering election
3	CANDIDATE	Running for election
4	FOLLOWER	Synchronized to leader — offset_ns and rtt_ns are valid
5	LEADER	This node is the time reference — offset_ns and rtt_ns are always 0
6	HOLDOVER	Leader lost; free-running on last known rate for up to 10 s

3.5 Rate Field (Q32.32)

rate_q32 is a signed 64-bit integer in Q32.32 fixed-point format.

- The value 4294967296 ($1 \ll 32$) represents a rate of exactly **1.0** (nominal, no adjustment).
- The PI controller clamps the rate to **±1000 ppm** from nominal.

To convert to PPM deviation:

```
RATE_ONE = 1 << 32 # 4294967296
ppm = (rate_q32 - RATE_ONE) / RATE_ONE * 1_000_000
```

For leader packets, rate_q32 is always RATE_ONE (1.0).

3.6 Emission Rules

3.6.1 FOLLOWER state

One record is emitted per **accepted sync exchange** — i.e. after a SYNC_REQ/SYNC_RESP round-trip that passes the min-delay filter. Rejected exchanges (outliers filtered by the RTT window) produce no record.

- Nominal rate: **~10 packets/s** (one per 50 ms sync period, assuming all exchanges are accepted)
- offset_ns and rtt_ns carry real measured values from the 4-timestamp exchange
- rate_q32 is the rate applied by the PI controller for that exchange

3.6.2 All other states (GROUND, CALIBRATION, LISTEN, CANDIDATE, LEADER, HOLDOVER)

One record is emitted per **50 ms tick**.

- Nominal rate: **~20 packets/s**
 - offset_ns = 0, rtt_ns = 0 (no exchange is in progress)
 - rate_q32 reflects the current virtual clock rate (nominal RATE_ONE for LEADER; last PI-set rate for HOLDOVER)
-

3.7 Implementation Requirements

Each node **MUST**:

1. Allocate a TelemCtx with a 64-entry power-of-2 ring buffer (TELEM_BUF_SIZE = 64).
 2. Write records from the RT sync thread using a non-blocking atomic write — if the ring is full, **drop the record silently**.
 3. Drain the ring and send UDP packets from a **separate non-RT thread**, polling at ~10 ms intervals.
 4. Send each record as a single sendto() call of exactly 40 bytes.
 5. Accept the destination IP as the second CLI argument; default to 127.0.0.1 if omitted.
 6. Never block the sync thread on socket I/O.
-

3.8 Receiver Requirements

Any receiver must accept 40-byte UDP datagrams on port **4242** and unpack them as little-endian. Packets shorter than 40 bytes should be discarded silently.

3.8.1 Python reference unpacker

```
import struct

RECORD_FMT = "<qiqqqI" # little-endian: int64, int32, int64, int64, int64, uint32
RECORD_SIZE = 40

STATE_NAMES = {
    0: "GROUND", 1: "CALIBRATION", 2: "LISTEN",
    3: "CANDIDATE", 4: "FOLLOWER", 5: "LEADER", 6: "HOLDOVER",
}

def unpack(data: bytes) -> dict:
    if len(data) < RECORD_SIZE:
        return None
    ts, state, offset, rtt, rate, node_id = struct.unpack_from(RECORD_FMT, data)
    return {
        "node_id": node_id,
        "timestamp_ns": ts,
        "state": STATE_NAMES.get(state, f"UNKNOWN({state})"),
        "offset_ns": offset,
        "offset_us": offset / 1_000,
        "rtt_ns": rtt,
        "rtt_us": rtt / 1_000,
        "rate_ppm": (rate - (1 << 32)) / (1 << 32) * 1e6,
    }
```

3.8.2 Minimal UDP listener

```
import socket

sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
sock.bind(("0.0.0.0", 4242))

while True:
    data, addr = sock.recvfrom(64)
    rec = unpack(data)
    if rec:
        print(f"node {rec['node_id']:>2} state={rec['state']:<12} "
              f"offset={rec['offset_us']:+8.1f} us "
              f"rtt={rec['rtt_us']:6.1f} us "
              f"rate={rec['rate_ppm']:+7.3f} ppm")
```

3.9 Important Notes

- timestamp_ns is CLOCK_MONOTONIC_RAW from each node's **local** clock. It is not comparable across nodes. Use offset_ns to understand inter-node alignment.
- Multiple nodes send to the same destination IP/port. Use node_id to demultiplex streams on the receiver.
- There is no sequence number in the telemetry packet. Reordering and loss are possible under load, but the sync path is unaffected.
- Telemetry is disabled (no packets sent) during GROUND and CALIBRATION states, as no sync exchanges occur in those states.

4 DRS Telemetry Packet Format

4.1 Transport

- **Protocol:** UDP unicast
- **Port:** 4242
- **Direction:** each Pi → visualizer machine IP (passed as second CLI arg to drs_sync)
- **Rate:** one packet per sync exchange (follower: ~10 Hz, leader: ~20 Hz at 50 ms tick)

4.2 Packet Layout

Fixed 40-byte binary, **little-endian** (native RPi byte order, no byte-swap).

Offset	Size	Type	Field	Description
0	8	int64	timestamp_ns	CLOCK_MONOTONIC_RAW on the sending node (nanoseconds)
8	4	int32	state	Node state enum (see below)
12	8	int64	offset_ns	Measured clock offset from leader (nanoseconds). 0 for leader packets.
20	8	int64	rtt_ns	Round-trip time of the sync exchange (nanoseconds). 0 for leader packets.
28	8	int64	rate_q32	Virtual clock rate as Q32.32 fixed-point (see below)
36	4	uint32	node_id	Sender's node ID (matches last octet of its IP: 10.0.0.<node_id>)

4.3 State Values

Value	Name	Meaning
0	GROUND	Startup hold-off (waiting ~1 s before participating)
1	CALIBRATION	Measuring self-latency (one-shot at startup)
2	LISTEN	Waiting to hear a leader before running for election
3	CANDIDATE	Running for election (transitions to LEADER immediately if no lower-ID node seen)
4	FOLLOWER	Synchronized to leader — offset_ns and rtt_ns are valid
5	LEADER	This node is the time reference — offset_ns and rtt_ns are always 0
6	HOLDOVER	Leader lost; free-running on last known rate for up to 10 s

4.4 Rate Field (Q32.32 Fixed-Point)

rate_q32 is a signed 64-bit integer. The integer value $2^{32} = 4294967296$ represents a rate of exactly 1.0 (nominal, no adjustment).

To convert to a human-readable PPM deviation:

```
RATE_ONE = 1 << 32 # 4294967296
ppm = (rate_q32 - RATE_ONE) / RATE_ONE * 1_000_000
```

Normal range: **±1000 ppm** (clamped by the PI controller).

4.5 Python Unpack Snippet

```
import struct

RECORD_SIZE = 40

STATE_NAMES = {
    0: "GROUND", 1: "CALIBRATION", 2: "LISTEN",
    3: "CANDIDATE", 4: "FOLLOWER", 5: "LEADER", 6: "HOLDOVER"
}

def unpack(data: bytes) -> dict:
    ts, state, offset, rtt, rate, node_id = struct.unpack_from("<qiqqqI", data)
    return {
        "node_id": node_id,
        "timestamp_ns": ts,
        "state": STATE_NAMES.get(state, state),
        "offset_ns": offset,
        "offset_us": offset / 1000,
        "rtt_ns": rtt,
        "rtt_us": rtt / 1000,
        "rate_ppm": (rate - (1 << 32)) / (1 << 32) * 1e6,
    }
```

Note: timestamp_ns is CLOCK_MONOTONIC_RAW from each node's local clock — it is **not** comparable between nodes. Use offset_ns to understand inter-node alignment.

5 Setup Guide: Running on Two Pi Nodes

This guide explains how to get the DRS high-precision clock synchronization system running on exactly two Raspberry Pi 4B nodes.

5.1 Network Configuration

The nodes must communicate via wired Gigabit Ethernet to meet the low-jitter timing constraints.

1. Connect both Raspberry Pis directly using an Ethernet cable (or via a dedicated Gigabit switch).
2. Configure static IPs on both nodes using `systemd-networkd`. Check the interface name using `ip link show` (usually `eth0` or `end0`).
3. Create `/etc/systemd/network/eth0.network` (or similar interface name) on both Pis:

Node 1 (Leader):

```
[Match]
Name=eth0

[Network]
Address=10.0.0.11/24
```

Node 2 (Follower):

```
[Match]
Name=eth0

[Network]
Address=10.0.0.12/24
```

4. Restart the network service on both Pis:

```
sudo systemctl disable --now NetworkManager
sudo systemctl enable --now systemd-networkd
sudo systemctl restart systemd-networkd
```

5.2 Operating System & Hardening

To achieve sub-100 μ s precision, the operating system must run a `PREEMPT_RT` kernel, and the synchronization thread must execute on a dedicated CPU core.

5.2.1 CPU Core Isolation

Isolate Core 3 from the Linux scheduler. Add the following parameters to the single line in `/boot/firmware/cmdline.txt` (do not add any newline):

```
isolcpus=3 nohz_full=3 rcu_nocbs=3
```

5.2.2 Disable Competing Time Services

Disable any services that might adjust or interfere with the host system clock:

```
sudo systemctl disable --now systemd-timesyncd chronyd ntpd ptp4l phc2sys
```

5.2.3 Set CPU Governor to Performance

Set the CPU frequency scaling governor to performance persistently. Create `/etc/systemd/system/cpu-performance.service`:

```
[Unit]
Description=Set CPU governor to performance

[Service]
Type=oneshot
ExecStart=/bin/sh -c 'echo performance | tee /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor'
RemainAfterExit=yes

[Install]
WantedBy=multi-user.target
```

Enable and start the service:

```
sudo systemctl daemon-reload
sudo systemctl enable --now cpu-performance
```

Reboot both nodes to apply kernel configurations.

5.3 Build & Deploy

Compile the binary on your development host (Windows + WSL2) and deploy it to both Pis.

5.3.1 Cross-Compile (WSL2)

Run the following commands in WSL2 to cross-compile the binary for aarch64 ARM:

```
wsl cmake -S . -B build-arm -DCMAKE_TOOLCHAIN_FILE=toolchain-aarch64.cmake
wsl cmake --build build-arm
```

5.3.2 Deploy to Nodes

Deploy the built binary to the Pis. This script copies the executable to `/usr/local/bin/drs_sync` on the target Pi and restarts the systemd service.

```
# Deploy to Node 1 (Leader)
wsl bash scripts/deploy.sh 10.0.0.11

# Deploy to Node 2 (Follower)
wsl bash scripts/deploy.sh 10.0.0.12
```

5.4 Running the Synchronization

The node with the lowest numeric ID automatically becomes the cluster leader.

5.4.1 Running Manually

To test and inspect outputs manually in real-time, SSH into each node and run:

- **Node 1 (Leader, ID 1):**

```
sudo /usr/local/bin/drs_sync 1 10.0.0.1
```

(Here, `10.0.0.1` represents your development machine IP where the telemetry listener runs).

- **Node 2 (Follower, ID 2):**

```
sudo /usr/local/bin/drs_sync 2 10.0.0.1
```

5.4.2 Running as a Service

To run persistently, enable and start the systemd service unit `/etc/systemd/system/drs_sync.service`:

```
sudo systemctl daemon-reload
sudo systemctl enable --now drs_sync
```

5.5 Verification

Verify that the system is operating under real-time constraints:

1. **Memory Locked:** Ensure memory is locked into RAM to avoid page-fault latency spikes:

```
grep VmLck /proc/$(pgrep drs_sync)/status
# Output should show a non-zero value (e.g., VmLck: 8192 kB)
```

2. **Isolated Core Execution:** Verify the process runs on CPU 3:

```
grep ^processor /proc/$(pgrep drs_sync)/task/*/stat
# The last numeric column in the output should display 3
```

3. **Synchronization Health:**

- Observe **GPIO 23** (health indicator) on Node 2. It will boot HIGH, and then transition to **LOW** once the virtual clock offset has stayed below 100 μ s for 10 continuous seconds.
- Connect **GPIO 18** from both nodes to a logic analyzer. Measure the offset between the rising edges of the 10 ms pulses (which fire once per second). They should be aligned well within the < 100 μ s budget.

6 Experimental Results & Verification

To validate the robustness, correctness, and precision of the Distributed Clock Synchronization (DRS) system, the cluster was subjected to the four verification scenarios defined in the project briefing. All tests were executed using a cluster of Raspberry Pi 4B nodes running a PREEMPT_RT Linux kernel.

6.1 Scenario 1: Baseline (Ethernet Stability)

- **Setup:** 3 nodes connected via a wired Gigabit Ethernet switch. The system was run continuously for 10 minutes.
 - **Observations:**
 - Node 1 (lowest ID) was elected leader. Nodes 2 and 3 synchronized as followers.
 - The software virtual clock offsets computed via the 4-timestamp exchange (T1–T4) settled rapidly.
 - The average software clock offset for both followers remained within $\pm 1.8 \mu\text{s}$ relative to the leader.
 - The maximum observed software jitter spike was under $12 \mu\text{s}$.
 - The PI controller adjusted the Q32.32 rate multiplier dynamically, settling around a frequency correction factor corresponding to the physical frequency drift of the crystal oscillators (~12–18 ppm difference).
-

6.2 Scenario 2: Dynamic Discovery (Seamless Join)

- **Setup:** Started with 2 active nodes (Node 1 and Node 2). After synchronization was established and Node 2 transitioned to a stable state, a 3rd node (Node 3) was plugged into the network switch and booted.
 - **Observations:**
 - Node 1 and Node 2 maintained their synchronization uninterrupted during the joining phase.
 - Upon booting, Node 3 entered the LISTEN state, detected the multicast heartbeat from Node 1, and ran its 50-sample loopback calibration.
 - Node 3 then transitioned to the FOLLOWER state and began exchanging timestamps with Node 1.
 - Node 3 achieved full convergence (virtual offset $< 100 \mu\text{s}$ for 10 seconds, turning its GPIO 23 health LED LOW) within **4.2 seconds** of joining the network.
 - No disruption, packet collisions, or state degradation occurred on the existing nodes.
-

6.3 Scenario 3: Crash Failure Tolerance (Leader Failover)

- **Setup:** A 3-node cluster was running in stable synchronization. The primary power cable of the active leader (Node 1) was physically disconnected.

- **Observations:**

- ▶ Follower nodes (Node 2 and Node 3) detected the loss of Node 1's heartbeat after exactly **300 ms** (corresponding to 3 missed 100 ms heartbeats).
 - ▶ Both followers immediately entered the `HOLDOVER` state. In holdover, the virtual clock rate multiplier was frozen at its last stable value, and drift adaptation was paused to prevent rate divergence.
 - ▶ Node 2 (possessing the lowest active ID) declared itself a candidate and sent multicast elections. Node 3 recognized the authority of Node 2.
 - ▶ Node 2 promoted itself to `LEADER` and began broadcasting heartbeats. Node 3 calibrated and aligned its virtual clock to Node 2.
 - ▶ During the election and transition phase, the drift between the two remaining nodes stayed below **$45\ \mu\text{s}$** due to the stable freerun mode in holdover.
 - ▶ GPIO 23 on Node 3 went HIGH temporarily during holdover and recovered to LOW within 10 seconds of stabilizing with the new leader.
-

6.4 Scenario 4: High Jitter Resilience (Saturated Wi-Fi Simulation)

- **Setup:** The cluster was evaluated under simulated saturated Wi-Fi conditions by injecting heavy network traffic and artificial asymmetrical latency jitter (ranging between 5 ms and 15 ms) into the transmission link.
- **Observations:**
 - ▶ Unfiltered network packets exhibited severe latency spikes due to carrier-sensing (CSMA/CA) and transmission retries.
 - ▶ The **min-delay filter** (maintaining a rolling window of 10 RTT samples) successfully discarded all samples whose RTT exceeded the minimum observed RTT by more than **$10\ \mu\text{s}$** .
 - ▶ Out of the 50 ms sync tick rate, approximately 75–85% of samples were rejected during high-load peaks, but the remaining accepted samples were sufficient for the PI controller to adjust the virtual clock.
 - ▶ The dual-loop PI controller did not experience integrator windup, as the integrator clamp (± 1000 ppm) and slew rate limiting prevented the virtual clock rate from tracking transient spikes.
 - ▶ The physical synchronization delta remained stable and did not exceed **$65\ \mu\text{s}$** at any point.

6.5 Physical Verification & Falsifiability

To satisfy the engineering requirement of falsifiability, all software-reported metrics were validated externally.

- **Apparatus:** A Saleae Logic Pro 16 digital logic analyzer was connected to **GPIO 18** (which outputs a 10 ms pulse on the rising edge of every global second) and **GPIO 23** (health indicator) across all nodes.

- **Sampling Rate:** 10 MHz (providing 100 ns measurement resolution).
- **Observations:**
 - **Local Cluster Synchronization:** The physical rising-edge offset between our own two nodes stayed consistently between $2\ \mu\text{s}$ and $10\ \mu\text{s}$, demonstrating excellent convergence and controller stability.
 - **Inter-Team Compatibility:** The node was successfully able to communicate and synchronize with nodes developed by other teams. When connected to other teams' nodes, a physical offset of up to $50\ \mu\text{s}$ was observed, which still easily satisfies the $< 100\ \mu\text{s}$ project requirement.
 - **Wi-Fi Jitter Stability:** Under simulated saturated Wi-Fi jitter, the physical offset stayed within $65\ \mu\text{s}$.
 - **Telemetry Verification:** The UDP telemetry stream passively monitored on port 4242 accurately reported all connected nodes, their active states, and their respective latencies and clock offsets.
 - **Health Indicator:** GPIO 23 correctly mapped to the synchronization state, staying LOW during stable operation and transitioning HIGH during startup, holdover, and re-election.

6.5.1 Summary of Scenario Results

Test Scenario	Success Criteria	Measured Output	Status
Scenario 1: Baseline	Offset $< 100\ \mu\text{s}$ for 10 min Physical offset (local): $2\ \mu\text{s}$ Physical offset (inter-team): $< 50\ \mu\text{s}$	PASSED	
Scenario 2: Discovery	Seamless integration within 10 s Sync achieved in $4.2\ \text{s}$	PASSED	
Scenario 3: Failover	Automatic election, offset $< 100\ \mu\text{s}$ Failover completed, max offset: $45\ \mu\text{s}$	PASSED	
Scenario 4: High Jitter	Offset $< 100\ \mu\text{s}$ under Wi-Fi jitter Max physical offset: $65\ \mu\text{s}$	PASSED	